

A Closed Recurrence and Stationary Atlas for a Two-State Reflected Markov Fluid

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Abstract

We classify a two-state Markov-modulated fluid reflected at zero. The sign of one stationary drift separates positive recurrence, null recurrence, and linear escape. In the stable chamber we reconstruct the unique invariant law, including its boundary atom, two interior densities, every workload moment, and the regulator rate. Vanishing switching and drift rates are classified by closed environmental classes. This round is the **drift-trichotomy owner**.

1 Model and sharp trichotomy

Let J jump $0 \rightarrow 1$ at rate a and $1 \rightarrow 0$ at rate b . The fluid slopes are $r_0 = -d$ and $r_1 = c$. For now $a, b, c, d > 0$, and

$$Y_t = X_0 + \int_0^t r_{J_s} ds, \quad L_t = \sup_{0 \leq s \leq t} (-Y_s)^+, \quad X_t = Y_t + L_t. \quad (1)$$

Thus L is the minimal right-continuous regulator and can increase only at $(X, J) = (0, 0)$. Put

$$\bar{r} = \frac{ac - bd}{a + b}. \quad (2)$$

Theorem 1 (Complete positive-core classification). *The formula (1) defines the unique global reflected PDMP. If $ac < bd$, it is positive recurrent with one invariant probability. If $ac = bd$, it is null recurrent and has no invariant probability. If $ac > bd$, it is transient and $X_t/t \rightarrow \bar{r} > 0$ almost surely.*

In the stable chamber let

$$\kappa = \frac{bd - ac}{cd}, \quad p_* = \frac{bd - ac}{(a + b)d}.$$

The only boundary atom is p_* at $(0, 0)$, and on $x > 0$ the state densities are

$$f_0(x) = \frac{ac\kappa}{(a + b)d} e^{-\kappa x}, \quad f_1(x) = \frac{a\kappa}{a + b} e^{-\kappa x}. \quad (3)$$

Proof. The Skorokhod map on the half-line proves existence, uniqueness, and nonnegativity. The environment has stationary law $\pi = (b, a)/(a + b)$, so its ergodic theorem gives $Y_t/t \rightarrow \bar{r}$.

Observe X at successive starts of state 1. If $I_n \sim \text{Exp}(b)$ and $O_n \sim \text{Exp}(a)$ are the following on- and off-times, then

$$W_{n+1} = \max\{0, W_n + cI_n - dO_n\}, \quad \mathbb{E}(cI_n - dO_n) = \frac{c}{b} - \frac{d}{a}. \quad (4)$$

The increments are continuous, nondegenerate, and have finite variance. The negative-mean Lindley chain is positive recurrent; at zero mean it is recurrent but has only an infinite invariant measure; at positive mean it escapes linearly. Renewal normalization by $\mathbb{E}(I_n + O_n) = 1/b + 1/a$ yields (2), including the asserted continuous-time speed.

For $ac < bd$, the stationary forward equations on $x > 0$ are

$$0 = df'_0 - af_0 + bf_1, \quad 0 = -cf'_1 + af_0 - bf_1. \quad (5)$$

Their sum and integrability give $df_0 = cf_1$. Substitution gives the decay rate κ . Boundary flux $ap_* = df_0(0)$ and normalization give (3). Irreducibility and positive recurrence give uniqueness. $\square$